Read a letter page

A model on your own laptop reads handwritten letters from the 1860s. You measure how wrong it is, write a better prompt, and try it on the next letter. Nothing leaves your laptop.

Workshop, path 1: LM Studio

How to work

Work through the steps in order.

There are no prompts to copy: you write your own.

What is in the folder

Download `letters.zip` from the workshop's repository and unzip it.

The folder `letters` holds three letter pages, numbered 1 to 3, each as a picture (`.jpg`) and as a transcription (`.txt`).

It also holds `cer.html`, a page that counts errors. Double-click it to open it in your browser; it works offline.

0 Does it fit?

Find out how much memory your laptop has.

Mac: Apple menu → *About This Mac*, the line *Memory*

Windows: *Settings* → *System* → *About*, the line *Installed RAM*

LM Studio needs a Mac with Apple Silicon and macOS 14 or newer, or a Windows laptop.

With an Intel Mac or less than 8 GB, pair up with someone whose laptop fits, or work on the notebooks instead.

0 Five sizes of Gemma 4

Gemma 4	memory	download size Mac	download size Windows
E2B	8 GB	4.3 GB	4.4 GB
E4B	16 GB	6.8 GB	6.3 GB
12B	16 GB	6.7 GB	7.6 GB
26B A4B	32 GB	15.6 GB	18.0 GB
31B	32 GB	18.4 GB	19.9 GB

Each model needs a laptop with at least the memory shown.

0 Which one?

Take the largest model your memory allows.

The E2B is the fastest but makes the most errors; it is the one for 8 GB.

The E4B is the usual choice with 16 GB. The 12B is a little more exact if you close other programs.

With 32 GB, the 26B A4B is nearly as exact as the 31B and much faster.

The 31B is the most exact and the slowest.

0 What the names mean

B stands for billions of parameters, the numbers a model computes with; the lecture's filter had nine. In the 12B and the 31B, all of them work on every word.

E stands for *effective*. The E2B and E4B work with 2.3 and 4.5 billion parameters, but have 5.1 and 8 billion with their lookup tables, so the download is larger than the name suggests.

A stands for *active*. In the 26B A4B, 8 of 128 experts work on each word, 3.8 billion parameters: it needs the memory of a 26B but runs nearly as fast as a 4B.

0 Mac: MLX. Windows: GGUF.

On a Mac, take *MLX*, Apple's own framework for its chips; it comes with LM Studio.

On Windows, take *GGUF*.

Take the *4-bit* version (in GGUF: Q4_K_M). It stores each number in 4 bits instead of 16, so it is smaller and a little less exact.

If a model will not load, update LM Studio.

0 Install

Install LM Studio from lmstudio.ai.

In *Model Search*, search for `gemma-4` and download the model you chose (Mac: MLX, 4-bit; Windows: GGUF).

In the *Chat* tab, load the model at the top and ask it a question.

1 First reading

Start a new chat and attach letter 1 (`1-joseph-1868-03-27.jpg`).

Ask, in your own words, for the text of the page.

Compare the reply with the scan.

2 Measure

Open `cer.html`.

Paste letter 1's transcription (the `.txt`) into *Reference* and the model's text into *Model output*.

The page shows the character error rate (CER) and marks every difference.

3 Refine the prompt

Look at what the model got wrong and add an instruction to your prompt that would have prevented it.

Ask again with the new prompt in a new chat, so that the model does not see its first answer, and measure again.

Repeat this two or three times.

4 Another letter

Try your best prompt, unchanged, on letter 2 and then on letter 3, each in a new chat, and measure both.

Does the prompt still work, or did it only suit letter 1?

Change it until one prompt works on all three letters, and measure all three again.

5 Beyond transcription

Ask for something other than the text, on any of the letters.

For example, ask for a list of the people, places and dates, and check each against the transcription. What did the model invent?

Or ask for a summary in two sentences, or a Swedish translation. What is missing, and what was added?

6 Your own material

Take a page of your own and try your best prompt on it.

What is different about this page, and did the model notice?